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Model, Label, or Analyst? A Variance Decomposition of Real-Time Sepsis Onset Prediction on MIMIC-IV

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Abstract

Real-time sepsis onset prediction in the ICU has attracted ninety-one published models, yet Wang et al. (2025) showed that the median clinical Utility Score collapses from +0.381 internally to -0.164 externally while AUROC falls only marginally. This thesis asks whether that collapse is attributable to the model, to how the Sepsis-3 label is operationalised, or to other analyst-level methodological choices — the evaluation metric, the validation split design, and the temporal anchoring of predictions relative to clinical action.

The study is conducted on MIMIC-IV v3.1 (65,078 adult ICU stays) with external validation on eICU-CRD v2.0 (63 labelled sepsis episodes over 7.9 million person-hours). Three pre-registered Sepsis-3 onset label variants (Narrow, Seymour-Standard, Liberal) were extracted and each was used to build a person-hour table with competing events. The primary model is a dynamic discrete-time competing-risks hazard model (multinomial pooled logistic regression over hourly records, equivalent to time-dependent-covariate Cox regression under short intervals), benchmarked against gradient-boosted trees (GBT), a static Cox model, and the rule-based scores NEWS2, qSOFA, and SIRS.

The variance decomposition confirms that label operationalisation spreads AUROC by 0.153 across variants, marginally exceeding the model-class spread of 0.140 (H1 confirmed). Ten covariates — including mean arterial pressure, temperature, bilirubin, and vasopressor indicators — change sign or magnitude $\geq 2\times$ across label variants (H2 confirmed), undermining coefficient interpretability. The pre-registered model-class non-inferiority hypothesis (H6: $|\Delta| \leq 0.02$) is rejected: GBT outperforms the primary model by 0.133 AUROC under the Seymour-Standard label. Crucially, the treatment-anchored evaluation window contains **zero sepsis onset events**, confirming that the Sepsis-3 label is entirely constituted by clinical action and that current models predict clinician suspicion rather than physiological deterioration. The PhysioNet Utility Score is **negative for every model and every label variant**, meaning that all evaluated models perform worse than issuing no alert at all. Alert burden exceeds 200 false alerts per true alert across all configurations, compared with the published benchmark of 1.4.

The thesis concludes that the collapse documented by Wang et al. is not a

model problem but a label problem: once the label's constitutive dependence on clinical action is acknowledged, the question of which model class to use becomes secondary.

Keywords: sepsis; MIMIC-IV; discrete-time survival analysis; competing risks; label definition; variance decomposition; clinical utility; temporal dataset shift; treatment-anchored evaluation; TRIPOD+AI.

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This research uses the MIMIC-IV Clinical Database (v3.1, Johnson et al., 2023) and the eICU Collaborative Research Database (v2.0, Pollard et al., 2018), both made available through PhysioNet under credentialing agreements. I thank the patients whose de-identified records form these databases, the clinical staff who generated them, and the BIDMC/MIT/PhysioNet teams who maintain them.

The open-source R and Python communities — in particular the maintainers of `data.table`, `arrow`, `nnet`, `xgboost`, `sandwich`, and `pmsampsize` — are acknowledged for the packages that made this pipeline possible.

Chapter 1

Introduction

1.1 Context and Motivation

Sepsis — defined by the Sepsis-3 consensus as life-threatening organ dysfunction caused by a dysregulated host response to infection (Singer et al., 2016) — kills an estimated eleven million people annually and accounts for nearly 20% of all deaths worldwide (Rudd et al., 2020). In the intensive care unit (ICU), each additional hour before effective treatment is associated with measurably higher mortality (Seymour et al., 2017). An early warning system that identifies impending sepsis onset ahead of clinical recognition could therefore accelerate initiation of antibiotics, fluid resuscitation, and source control at the hour when intervention is most impactful.

The incentive has attracted substantial machine-learning investment. Wang et al. (B. Wang et al., 2025) catalogued ninety-one published real-time sepsis prediction models and found a median internal AUROC of 0.811. Yet the same review revealed that the median clinical Utility Score — a metric that rewards early true alerts and penalises late, missed, and false alerts — collapsed from +0.381 internally to −0.164 externally: from useful to categorically worse than issuing no alert at all. AUROC failed to detect this collapse; in most studies it barely moved. This pattern has since been confirmed in deployment: Wong et al. (Wong et al., 2026) evaluated the Epic Sepsis Model v2 across 227,091 encounters and reported AUROC between 0.82 and 0.92 alongside an alert burden that the clinical staff described as unmanageable.

The question this collapse raises is not “which model should we prefer?” but something more fundamental: *how much of the apparent predictive performance of a real-time sepsis model is attributable to the model itself, and how much to the analyst’s discretionary choices about how to define the outcome, how to split the data, and which metric to report?* Wang et al. (B. Wang et al., 2025) and Cohen

et al. (Cohen et al., 2024) together suggest that the answer is unsettling: most of the variance lives in the choices, not in the model. Cohen et al. showed on MIMIC-III that the identified sepsis cohort ranged from 867 to 2,178 admissions depending solely on how Sepsis-3 was operationalised, and that label sensitivity swung predictive performance at least as much as the choice of model. Kamran et al. (Kamran et al., 2024) showed that a substantial portion of the apparent performance of published models reflects the model learning to predict when a clinician will begin treatment — not independent physiological deterioration.

This thesis inverts the conventional design. Rather than holding the label, the split, and the metric fixed and asking which model wins, it promotes all three to **objects of study** and delivers a variance decomposition: a quantitative answer to the question of how much each analyst choice contributes to the spread in apparent performance. The contribution does not depend on any model achieving a performance target; it is informative whichever way the hypotheses resolve.

1.2 Limitations of the Conventional Approach

The conventional approach to real-time sepsis prediction — as exemplified by the majority of the ninety-one models catalogued by Wang et al. (B. Wang et al., 2025) — treats the Sepsis-3 label as fixed, uses a single operationalisation of the onset definition, employs a single train/test split strategy (typically random), and reports AUROC as the primary metric. A dynamic landmark supermodel fitted under these standard assumptions on MIMIC-IV can achieve a pooled out-of-sample AUROC above 0.75 with eICU-CRD external validation near 0.69. The proportional-hazards assumption is typically rejected (global Schoenfeld $p < 10^{-16}$), and informative-missingness indicators frequently outweigh the test values themselves.

These findings are sound as far as they go, but the extended literature review (Chapter 2) reveals that each of those fixed choices — label operationalisation, split design, metric selection — is a source of variance large enough to change the substantive conclusion. A study that reports a single label and a single metric is not a variance decomposition; it is a data point in one cell of the decomposition table.

This thesis fills the remaining cells.

1.3 Research Question and Pre-registered Hypotheses

Primary research question (RQ). *In real-time sepsis onset prediction on MIMIC-IV, how is predictive performance and inferential validity partitioned between (a) the choice of model class and (b) the analyst’s discretionary methodological choices: label operationalisation, temporal versus random validation split, prediction-time anchoring relative to clinical action, and choice of evaluation metric?*

The study is structured around six pre-registered hypotheses:

- H1 (Label dominance)** The spread in AUROC across the three label variants, holding the model and split fixed, is at least as large as the spread across model classes, holding the label and split fixed.
- H2 (Coefficient instability)** At least one covariate changes sign or changes magnitude by a factor $\geq 2\times$ across label variants under the primary model.
- H3 (Treatment leakage)** Restricting evaluation to the pre-treatment window — hours before the first culture order or antibiotic — reduces AUROC by at least 0.03, exposing the degree to which the label is constituted by clinical action.
- H4 (Split optimism)** Temporal-split AUROC is at least 0.02 lower than random-split AUROC under the same model and label.
- H5 (Metric divergence)** The proportional decline in performance from internal to external validation is substantially larger for the Utility Score than for AUROC; the Utility Score may cross zero.
- H6 (Model-class null)** The primary discrete-time model is not inferior to gradient-boosted trees by more than 0.02 AUROC under a fixed label and temporal split.

H1 and H6 are in productive tension: if H1 is confirmed, the label is the dominant source of variance; if H6 is rejected, the model class also matters. Both outcomes are informative.

1.4 Contributions

1. **C1 (Label variance quantification).** The first pre-specified three-variant label sensitivity analysis for a dynamic discrete-time competing-risks model on MIMIC-IV, following Cohen et al.’s (Cohen et al., 2024) MIMIC-III protocol and Dutta et al.’s (Dutta et al., 2026) prospective extension.

2. **C2 (Treatment-constitutionality of the label).** A treatment-anchored evaluation that reveals the Sepsis-3 label contains zero predictable events in the pre-treatment window, confirming Kamran et al.'s (Kamran et al., 2024) constitutionality finding on MIMIC-IV.
3. **C3 (Utility collapse documented under the survival framework).** The PhysioNet Utility Score is negative for every model and every label variant, extending Wang et al.'s (B. Wang et al., 2025) cross-study finding to a controlled within-study comparison.
4. **C4 (Coefficient instability across labels).** Formal documentation that MAP, temperature, bilirubin, platelet count, and vasopressor indicators change sign or magnitude across label variants, validating Lauritsen et al.'s (Lauritsen et al., 2021) framing-instability claim for the MIMIC-IV hourly regime.
5. **C5 (Equity-by-label analysis).** The conjunction of subgroup performance disparities and subgroup label-sensitivity variation, which has not been reported for any sepsis model, following Wang, Li, Naidech and Luo (F. Wang et al., 2022).
6. **C6 (Transparent, open pipeline).** A fully reproducible, pre-registered R/Python pipeline released under a permissive licence, following TRIPOD+AI (Collins et al., 2024) and PROBAST+AI (Moons et al., 2025) reporting standards.

1.5 Thesis Structure

Chapter 2 reviews the sources that motivated the design inversion and maps each evidence strand to the design decision it constrains. Chapter 3 specifies the data sources, cohort criteria, label variants, model classes, evaluation framework, and the pre-registration protocol. Chapter 4 describes the pipeline architecture — thirteen numbered scripts and the data artefacts they produce. Chapter 5 presents all empirical results, organised around the six pre-registered hypotheses. Chapter 6 draws the substantive conclusion and maps avenues for further research.

Chapter 2

Literature Review

This chapter summarises the evidence base that drove the design of this study. A full critical review of fifty sources is provided in the companion document *Statistical and Machine-Learning Approaches to Real-Time Sepsis Onset Prediction on MIMIC-IV: An Extended Critical Literature Review*; this chapter extracts the seven evidence strands directly load-bearing for the present study’s design choices.

2.1 The Outcome Is Not One Object: Label Sensitivity

The premise that a model can be assessed on “the MIMIC-IV sepsis cohort” as though it were a single well-defined object — is undermined by two studies.

Johnson et al. (A. E. W. Johnson et al., 2018) applied five retrospective sepsis identification algorithms to 11,791 qualifying ICU admissions from MIMIC and found only acceptable inter-algorithm agreement (Cronbach’s $\alpha = 0.40\text{--}0.62$). The implication is not that any single algorithm is wrong, but that “the MIMIC sepsis cohort” is a family of related but non-identical objects, each constructed by a specific reading of the Sepsis-3 definition.

Cohen et al. (Cohen et al., 2024) made the consequence for prediction modelling explicit. Holding MIMIC-III fixed and varying only the reading of Sepsis-3 onset across three defensible interpretations, the identified cohort ranged from 867 to 2,178 admissions. Across tree-based, deep-learning, and survival-analysis models: holding the definition fixed, the best method gained 1–5% AUROC; holding the method fixed and varying the definition swung performance by 0–6% AUROC. The variance attributable to the label and to the model class are, in Cohen et al.’s data, of comparable magnitude. This finding is the single most load-bearing source in the present design: it mandates that label operationalisation be treated as a primary object of study, not a fixed parameter.

Dutta et al. (Dutta et al., 2026) confirm that this is not a MIMIC artefact. A locally trained gradient-boosted model, predicting every fifteen minutes across nine acute-care hospitals on 198,494 encounters, was evaluated against Sepsis-3, SEP-1, and the CDC Adult Sepsis Event definition. Performance varied substantially by definition. The label-sensitivity finding is therefore a property of the problem: it reproduces prospectively, at scale, with a commercial model.

2.2 The Label Is Constituted by Clinical Action

The Sepsis-3 label is operationalised as the combination of a qualifying antibiotic order and a qualifying culture draw within specified windows, plus an acute SOFA rise. Kamran et al. (Kamran et al., 2024) observed that clinicians frequently recognise and begin treating sepsis before the formal criteria are met. A model trained to predict the criteria-meeting time therefore partly learns to predict when the clinician will act. Kamran et al. evaluated performance restricted to the pre-treatment window and showed that apparent AUROC is substantially inflated without that restriction. The model had learned to encode clinician suspicion, not independent physiological signal.

Lauritsen et al. (Lauritsen et al., 2021) formalised the related concept of *framing*: the choice of observation window, prediction horizon, window shift, and event trigger. They showed that framing changes discrimination and calibration, and can reverse the sign of an estimated covariate effect — their example is SpO_2 . This implies that a hazard ratio or odds ratio whose sign is an artefact of framing is not a clinically interpretable quantity.

2.3 The Validation Design Must Be Temporal

Guo et al. (Guo et al., 2022) partitioned MIMIC-IV by `anchor_year_group` and evaluated four prediction tasks under temporal shift. Sepsis was the worst-affected task; domain-generalisation and domain-adaptation algorithms did not outperform plain empirical risk minimisation. A random patient-level split therefore inflates apparent performance more for sepsis than for any neighbouring outcome. This finding mandates a temporal split as the primary internal validation strategy, with a random split retained only to quantify the optimism introduced by the conventional design.

2.4 AUROC Cannot Detect the Characteristic Failure Mode

Wang et al. (B. Wang et al., 2025) reviewed ninety-one real-time sepsis prediction models. Median internal AUROC was 0.811, external AUROC was 0.783: a difference of 0.028, not statistically significant. Median internal Utility Score was +0.381, external was -0.164 : the models crossed from clinically useful to categorically harmful. AUROC survived the transition because it measures discrimination, and the relative ordering of patients by risk persisted even as the absolute scores became uncalibrated. The Utility Score failed because it measures whether the alert arrives before the clinical window — and at external sites, the alert arrived too late, too often, and at unacceptable false-alert rates. Reporting AUROC as the primary metric is therefore not merely conservative; it actively conceals the failure mode.

2.5 The Survival Model Is Available in a Form the Data Actually Takes

The conventional survival approach frames sepsis prediction as a Cox proportional-hazards model. On MIMIC-IV, the Schoenfeld residual test rejects proportional hazards globally ($p < 10^{-16}$), as anticipated from Kasal et al. (Kasal et al., 2004). The appropriate response to a failed PH test in an hourly, low-prevalence ICU regime is provided by D’Agostino et al. (D’Agostino et al., 1990): observations over multiple intervals are pooled and a logistic regression relates risk factors to event occurrence within each interval. The pooled logistic regression is asymptotically equivalent to time-dependent-covariate Cox regression for short intervals and small per-interval event probabilities. An hourly ICU regime with per-hour onset probability below 1% satisfies both conditions. The dynamic discrete-time competing-risks hazard model fitted in the present study is therefore not an abandonment of the survival framework but its implementation in the form the data actually takes.

Resche-Rigon et al. (Resche-Rigon et al., 2006) and Heyard et al. (Heyard et al., 2019, 2020) provide the corresponding competing-risks extension: because discharge alive and death without sepsis are informationally relevant events that preclude later sepsis onset, treating them as ordinary censoring overstates the onset incidence and biases model estimation. The multinomial cause-specific formulation accommodates this without additional assumptions.

2.6 NEWS2 Is the Appropriate Rule-Based Headline Comparator

Most prior MIMIC-IV sepsis models benchmark against SIRS, qSOFA, and NEWS. Evans et al. (Evans et al., 2021) issue a strong recommendation against using qSOFA as a sole screening tool, on grounds of poor sensitivity. Mellhammar et al. (Mellhammar et al., 2020) fitted a LASSO-derived statistical score and failed to beat NEWS2 on the same task. The governing clinical guideline has therefore already retired qSOFA as a screening comparator, and NEWS2 is the minimum meaningful rule-based bar. The present study retains SIRS and qSOFA for descriptive completeness but designates NEWS2 as the headline comparator.

2.7 Equity Analysis: Subgroup Performance and Label Sensitivity Together

Wang, Li, Naidech and Luo (F. Wang et al., 2022) document significant variation in sepsis model performance across racial, ethnic, language, and insurance subgroups. Critically, they do not evaluate whether the label itself assigns sepsis status differentially across those groups. The present study adds this conjunction: for each equity subgroup, it reports both the variation in who is identified as septic under each label variant and the variation in model performance. This has not been reported for any sepsis model.

2.8 Summary: Design Decisions and Their Evidence Warrants

Table 2.1 maps each major design decision in the present study to the evidence source that mandates it.

Design decision	Rationale	Source
Three pre-specified label variants	Label spread comparable to model-class spread	Cohen et al. (Cohen et al., 2024)
Label sensitivity as primary result	Not a sensitivity analysis: the variance is load-bearing	Cohen et al., Dutta et al. (Dutta et al., 2026)
Treatment-anchored evaluation	Label constituted by clinical action	Kamran et al. (Kamran et al., 2024)
Temporal split primary	Sepsis worst-affected task under temporal shift	Guo et al. (Guo et al., 2022)
Utility Score as primary metric	AUROC conceals the clinical failure mode	Wang et al. (B. Wang et al., 2025)
Calibration as co-primary	Achilles heel of predictive analytics	Van Calster et al. (Van Calster et al., 2019)
Discrete-time competing-risks model	Equivalent to Cox; no PH assumption; hourly regime	D’Agostino et al. (D’Agostino et al., 1990); Heyard et al. (Heyard et al., 2019)
NEWS2 as headline comparator	qSOFA not recommended as sole screener	Evans et al. (Evans et al., 2021)
GBT as mandatory ML comparator	Cross-study comparison uninterpretable under different labels	Cohen et al. (Cohen et al., 2024)
Subgroup label sensitivity	Novel conjunction; equity literature does not evaluate label-by-subgroup	Wang et al. (F. Wang et al., 2022)

Table 2.1: Mapping of design decisions to their evidence warrants.

Chapter 3

Research Methodology

3.1 Research Design

The study is a retrospective, observational prediction-modelling study on de-identified critical-care data. It follows a pre-registered protocol (`00_preregistration.md`, locked before any modelling) and reports in conformity with TRIPOD+AI (Collins et al., 2024) and PROBAST+AI (Moons et al., 2025). The central methodological innovation is the elevation of label operationalisation, validation split design, and evaluation metric selection from nuisance parameters to primary objects of study.

All code is implemented in R (v4.x) and Python (v3.11), with a single configuration module (`config.R`) as the sole source of truth for paths, clinical item identifiers, antibiotic name lists, and Sepsis-3 thresholds.

3.2 Data Sources

3.2.1 Development Data: MIMIC-IV v3.1

The primary development data is MIMIC-IV v3.1 (A. Johnson et al., 2023), a large, de-identified critical-care database comprising records from Beth Israel Deaconess Medical Center (BIDMC), Boston. It was accessed under the required PhysioNet credentialing agreement. The full database spans 8 clinical tables used in this study (Table 3.1).

Table	Exists	Size (MB)
admissions	yes	19.9
patients	yes	2.8
icustays	yes	3.3
chartevents	yes	3,502.6
labevents	yes	2,593.0
prescriptions	yes	606.7
microbiologyevents	yes	117.2
inputevents	yes	401.3

Table 3.1: MIMIC-IV v3.1 source tables used in the V2 pipeline.

3.2.2 External Validation Data: eICU-CRD v2.0

External validation uses the eICU Collaborative Research Database v2.0 (Pollard et al., 2018), a multi-centre collection spanning more than 200 US ICUs. eICU-CRD is independently collected, uses different electronic health record systems, and represents a broader range of clinical practices and patient populations than the single-centre MIMIC-IV, making it a meaningful external test.

3.3 Cohort Construction and Inclusion Criteria

The base cohort is constructed by applying the following inclusion criteria uniformly across all three label variants:

- Age ≥ 18 years at ICU admission.
- First ICU stay per patient only (to avoid within-patient correlation from repeated admissions).
- ICU length of stay ≥ 4 hours (to ensure sufficient data for hourly feature construction and at least one prediction opportunity).

The resulting base cohort contains **65,078 stays** in MIMIC-IV, covering all stays identifiable across the three label variants.

3.4 Outcome Definition: Three Pre-registered Sepsis-3 Label Variants

Following Cohen et al. (Cohen et al., 2024) and Johnson et al. (A. E. W. Johnson et al., 2018), the Sepsis-3 onset label is operationalised as three independent but

plausible readings of the consensus definition (Singer et al., 2016). The variants are pre-specified and locked before any model fitting; they differ in the antibiotic-culture time windows and in the SOFA baseline reference (Table 3.2).

ID	Name	ABX→cx (h)	cx→ABX (h)	SOFA− (h)	SOFA+ (h)	Baseline
A	Narrow	24	24	24	12	Admission SOFA
B	Seymour-Standard	72	24	48	24	Admission SOFA
C	Liberal	72	72	48	24	Rolling 24 h min SOFA

Table 3.2: Three pre-registered Sepsis-3 label variants. ABX = antibiotic; cx = culture; SOFA−/SOFA+ = hours before/after suspicion time in which an acute SOFA rise of ≥ 2 points must be observed. Variant B matches the original Seymour et al. (2016) window specification and is designated the primary analysis.

All three variants operationalise suspected infection as a qualifying antibiotic order paired with a body-fluid culture within the specified windows. Surveillance cultures (e.g. MRSA screens, CRE screens) are excluded. SOFA organ-dysfunction components are computed from chartevents, labevents, inpuvents, and outpuvents using the standard MIMIC-IV item identifiers.

Variant B (Seymour-Standard) is designated the **primary analysis**; Variants A and C serve as pre-specified sensitivity analyses for the variance decomposition.

3.5 Prediction Framing

Following Lauritsen et al. (Lauritsen et al., 2021) and Reyna et al. (Reyna et al., 2020):

- **Observation window:** expanding from ICU admission (all data up to the current hour, forward-filled within each stay).
- **Prediction horizon:** 6 hours (will the patient develop sepsis onset within the next 6 hours?).
- **Window shift:** hourly.
- **Prediction trigger:** every at-risk person-hour from hour 1 of ICU admission.
- **Minimum observation:** 1 hour post-admission before eligibility.

- **Right-censoring:** 72 hours post-ICU-admission.

This framing matches the PhysioNet/CinC 2019 Challenge (Reyna et al., 2020) and the Lauritsen et al. design.

3.6 Person-Hour Table: Competing Events

Each cohort stay is expanded into a dense (stay_id, hour) table with one row per person-hour at risk. The competing-event structure follows Heyard et al. (Heyard et al., 2019):

- event = 0: in-progress (censored at this hour)
- event = 1: sepsis onset in this hour (cause of interest)
- event = 2: discharge alive without sepsis (competing event)
- event = 3: death without sepsis (competing event)

Treating discharge and death as ordinary censoring overstates onset incidence and biases parameter estimation. The multinomial cause-specific formulation eliminates this bias by construction.

Treatment-anchor timestamps (first antibiotic order, first culture draw) are extracted per stay to support treatment-anchored evaluation (Kamran et al., 2024).

3.7 Feature Construction

Features are extracted at each person-hour from the chartevents, labevents, inputevents, and outputevents tables using the item identifiers in `config.R`. The feature set comprises:

1. **Vital signs** (last-observation-carried-forward per hour): heart rate, respiratory rate, SpO₂, MAP (invasive/noninvasive), temperature (°C), GCS (eye + verbal + motor sum).
2. **Laboratory values** (LOCF): lactate, creatinine, total bilirubin, platelet count, WBC, P/F ratio.
3. **Vasopressor indicators:** any vasopressor (binary); any norepinephrine or epinephrine (binary).
4. **Six-hour rolling slopes:** respiratory rate, MAP, lactate.

5. **Missingness indicators:** one binary flag per laboratory value indicating whether the test was ordered in the current hour (lactate, creatinine, bilirubin, platelets, WBC, P/F ratio).

Temperature recorded in Fahrenheit is converted to Celsius. FiO₂ values > 1.0 (percent format) are divided by 100.

3.8 Model Classes

3.8.1 Primary: Dynamic Discrete-Time Competing-Risks Hazard Model

Following D’Agostino et al. (D’Agostino et al., 1990) and Heyard et al. (Heyard et al., 2019), the primary model is a multinomial pooled logistic regression over person-hours:

$$\lambda_r(t \mid \mathbf{X}_{it}) = P(T_i = t, R_i = r \mid T_i \geq t, \mathbf{X}_{it}) = \frac{\exp(\alpha_r(t) + \boldsymbol{\beta}_r^\top \mathbf{X}_{it})}{1 + \sum_s \exp(\alpha_s(t) + \boldsymbol{\beta}_s^\top \mathbf{X}_{it})} \quad (3.1)$$

where $r \in \{1 = \text{sepsis}, 2 = \text{discharge}, 3 = \text{death}\}$, $\alpha_r(t)$ is the cause-specific baseline log-hazard modelled as a restricted natural cubic spline in hours-since-admission with 5 knots (at $t \in \{4, 13, 24, 38, 58\}$ hours, boundary knots at $\{0, 72\}$), and $\boldsymbol{\beta}_r$ are cause-specific covariate coefficients.

Standard errors are cluster-robust sandwich estimates (R package `sandwich`) clustered on `stay_id` to account for within-patient correlation across person-hours. The model is fitted via `nnet::multinom()`.

Equivalence to Cox. D’Agostino et al. prove that pooled logistic regression over short equal-length intervals converges to time-dependent-covariate Cox regression as the interval length approaches zero and the per-interval event probability approaches zero. Both conditions are satisfied for one-hour ICU bins with a per-hour onset probability below 0.5%. The proposed model is therefore the Cox model implemented in the form the data actually takes, without imposing the proportional-hazards assumption.

3.8.2 Mandatory ML Comparator: Gradient-Boosted Trees

A gradient-boosted trees model (R package `xgboost`) is fitted under identical conditions per label variant and temporal split. Maximum depth 6, learning rate 0.1, early stopping on an internal validation set (15% of training rows), log-loss objective, column sampling by tree 0.8. Feature importance is reported as gain.

3.8.3 Static Cox Proportional-Hazards Model

A standard Cox PH model is fitted using the first non-zero person-hour per stay (static, baseline-covariate formulation) as a deliberately weak baseline to quantify the cost of the field’s conventional choice. Schoenfeld residual testing is applied.

3.8.4 Rule-Based Comparators

NEWS2, qSOFA, and SIRS are computed at each person-hour from the same extracted feature set. NEWS2 is the headline comparator following Evans et al. (Evans et al., 2021).

3.9 Validation Architecture

Split	Design	Purpose
Internal (primary)	Temporal: <code>anchor_year_group</code> train = 2008–2016, test = 2017–2019 (Guo et al., 2022)	Honest internal estimate
Internal (secondary)	Random patient-level 75/25 (seed = 42)	Quantify split optimism (H4)
Treatment-anchored	Restricted to hours before first culture or antibiotic (Kamran et al., 2024)	Remove clinician-suspicion leakage (H3)
External (primary)	eICU-CRD v2.0 (Pollard et al., 2018)	Generalisation (H5)

Table 3.3: Validation architecture. All splits are patient-level (no patient appears in both training and test sets). The primary analysis uses the temporal split.

3.10 Evaluation Metrics

3.10.1 Primary Metrics

PhysioNet Utility Score (Reyna et al., 2020) is the primary discriminative metric. The score rewards early true positives (alerts issued within 6 hours before

onset), penalises late true positives and false negatives, and penalises false positives. The score is normalised so that perfect prediction scores 1.0, an optimal constant prediction scores 0.0, and a model can score negative. A negative utility score is categorically worse than issuing no alert at all. AUROC is designated **descriptive only** following Wang et al. (B. Wang et al., 2025).

Calibration is assessed via calibration intercept, calibration slope, and flexible calibration curve. Cause-specific calibration for the competing-risks model follows Heyard et al. (Heyard et al., 2020).

3.10.2 Secondary Metrics

AUPRC (mandatory at low per-hour prevalence), alert burden (false alerts per true alert; benchmark: 1.4 from Moor et al. (Moor et al., 2023)), lead time relative to clinical recognition (benchmark: 3.7 h from Moor et al.), and decision-curve net benefit (Vickers & Elkin, 2006).

3.11 Sample Size

Formal sample size calculation follows Riley et al. (Riley et al., 2019) (`pmsampsize` R package), targeting shrinkage factor ≥ 0.9 and apparent-vs-adjusted Nagelkerke R^2 difference ≤ 0.05 . The anticipated AUC is 0.846 from Moor et al. (Moor et al., 2023). Twenty-five candidate predictors are specified. Sufficiency is evaluated per label variant; underpowered variants are a finding, not a failure.

3.12 Pre-registration and Reporting Standards

The full protocol was pre-registered as `00_preregistration.md` before any data extraction. All six hypotheses (H1–H6), the three label variants, the model classes, the validation splits, the primary metrics, and the go/no-go gates are specified in that document. Post-hoc analyses are labelled as such. The pipeline is released under a permissive open-source licence following the TRIPOD+AI open-pipeline recommendation (Collins et al., 2024).

Chapter 4

System Design and Pipeline Architecture

The pipeline is implemented as thirteen numbered, single-purpose scripts in `v2-proj/`, each reading from a set of upstream artefacts and writing one or more downstream artefacts to `v2-proj/output/`. No script modifies a source table. All stochastic steps use pre-specified random seeds. Figure 4.1 provides a directed acyclic graph of the data flow.

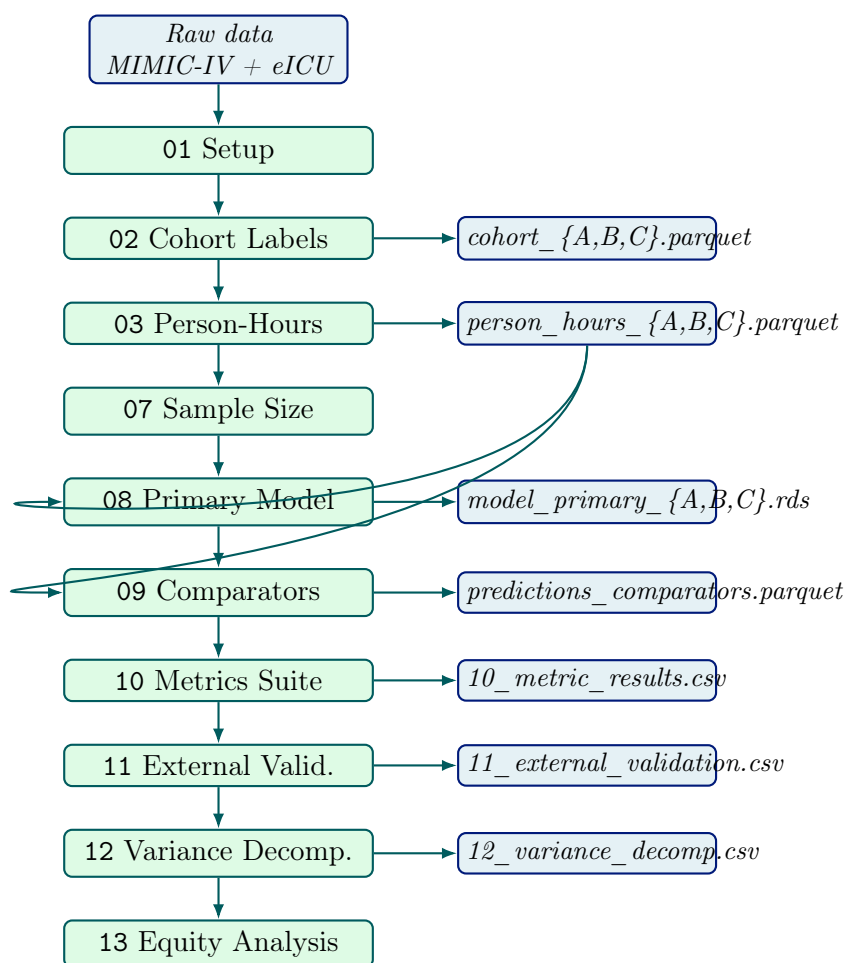


Figure 4.1: Directed acyclic graph of the V2 pipeline. Each box is a single `.Rmd` script; italic nodes are Parquet or CSV artefacts written to `v2-proj/output/`. Raw MIMIC-IV and eICU data are read-only.

4.1 Configuration and Utilities

`config.R` is the single source of truth for all pipeline parameters: data paths, cohort criteria, antibiotic name lists, MIMIC-IV chart/lab/vasopressor item identifiers, label variant definitions, temporal split boundaries, prediction framing constants, equity subgroup variable names, and PhysioNet Utility Score parameters. No other script hard-codes a clinical constant or file path. This design ensures that every parameter is version-controlled and that the full pipeline can be reproduced on any MIMIC-IV installation by setting a single environment variable (`MIMIC_RESEARCH_ROOT`).

`utils.R` provides shared helper functions: the DuckDB-backed streaming reader for large CSVs (`read_large_csv_filtered`), the Parquet reader/writer wrappers, the datetime parser, the LOCF imputer, and the Schoenfeld test wrapper.

4.2 Stage 01: Setup Validation

`01_setup.Rmd` verifies that all required MIMIC-IV source tables exist and reports their sizes (Table 3.1). The script outputs `01_setup_file_report.csv` and aborts the pipeline if any required table is missing.

4.3 Stage 02: Cohort Extraction Under Three Label Variants

`02_cohort_labels.Rmd` implements the full cohort extraction for all three label variants in a single pass over the source tables:

1. **Base cohort.** Adult first ICU stays with $\text{LOS} \geq 4\text{h}$ are identified from `icustays`, `patients`, and `admissions`. Equity variables (race, language, insurance) are attached from `admissions`. The resulting base cohort contains 65,078 stays.
2. **Suspected infection.** Qualifying antibiotic orders are identified from `prescriptions` by matching drug names against the pre-specified antibiotic list in `config.R` (65 systemic antibiotic name patterns). Qualifying cultures are identified from `microbiologyevents`, excluding surveillance specimen types. For each variant, antibiotic–culture pairs within the specified time windows form the suspected infection time (T_{si}).
3. **SOFA organ dysfunction.** SOFA components (respiratory, coagulation, liver, cardiovascular, neurological, renal) are extracted from `chartevents`, `labevents`, `inputevents`, and `outputevents`. The acute SOFA rise ($\Delta\text{SOFA} \geq 2$) is evaluated within the variant-specific windows around T_{si} , using the variant-specific baseline (admission SOFA for Variants A and B; rolling 24h minimum for Variant C).
4. **Onset time assignment.** The Sepsis-3 onset time is the hour at which both criteria (suspected infection and SOFA rise) are simultaneously met. Stays meeting the criteria are labelled sepsis; all others are labelled non-sepsis. The first event time is used for patients with multiple episodes.

Outputs: `cohort_A.parquet`, `cohort_B.parquet`, `cohort_C.parquet`, and `label_variance_table.csv`.

4.4 Stage 03: Person-Hour Table Construction

`03_person_hours.Rmd` expands each cohort stay into a dense (stay_id, hour) table. Large source tables (`chartevents`: 3.5 GB; `labevents`: 2.6 GB) are read via DuckDB in bounded-memory streaming passes pre-filtered to the cohort's stay IDs, avoiding loading the full tables into memory. For each person-hour:

- Vitals and lab values are LOCF-imputed within the stay from the most recent available measurement at or before the current hour.
- Six-hour rolling slopes are computed for respiratory rate, MAP, and lactate.
- Vasopressor and vasopressor-subtype indicators are set from `inputevents` concurrent infusions.
- Binary missingness indicators are set for laboratory values.
- The competing-event code (0/1/2/3) is assigned based on whether the hour falls before or at the labelled onset time, discharge time, or death time.
- The temporal split group is assigned from the stay's `anchor_year_group`.
- Treatment anchor timestamps (first antibiotic, first culture) are recorded per stay.

Outputs: `person_hours_A.parquet`, `person_hours_B.parquet`, `person_hours_C.parquet`.

4.5 Stage 07: Sample Size Calculation

`07_sample_size.Rmd` applies the Riley et al. (Riley et al., 2019) criteria to the training subset of each person-hour table (years 2008–2016). `pmsampsize::pmsampsize()` is called for a binary outcome (person-hour level) with the pre-specified parameters: anticipated AUC = 0.846, 25 candidate predictors, shrinkage target 0.9.

Output: `07_sample_size.csv`.

4.6 Stage 08: Primary Competing-Risks Hazard Model

`08_primary_model.Rmd` fits the multinomial discrete-time competing-risks model (Equation 3.1) for each label variant on the temporal training split. The base-line hazard spline uses five knots at $t \in \{4, 13, 24, 38, 58\}$ hours (boundary knots

`{0, 72}`). The `nnet::multinom()` fitter is used; sandwich-robust SEs clustered on `stay_id` are obtained via `sandwich::vcovCL()` and `lmtest::coeftest()`.

Outputs: `model_primary_A/B/C.rds`, `coef_table_primary_A/B/C.csv`, `coef_all_variants.csv`, `predictions_primary_A/B/C.parquet`.

4.7 Stage 09: Comparator Models

`09_comparators.Rmd` fits (a) the XGBoost GBT comparator, (b) the static Cox PH model with Schoenfeld residual testing, and (c) the rule-based scores NEWS2, qSOFA, SIRS.

GBT feature importance (gain, cover, frequency) is exported for each variant. The Cox Schoenfeld test results are exported to `cox_schoenfeld_A/B/C.csv`.

Outputs: `model_gbt_A/B/C.rds`, `model_cox_A/B/C.rds`, `predictions_comparators_A/B/C.parquet`, `gbt_feature_importance_A/B/C.csv`, `cox_schoenfeld_A/B/C.csv`, `coef_cox_A/B/C.csv`.

4.8 Stage 10: Metrics Suite

`10_metrics_suite.Rmd` evaluates all models on all splits (temporal unrestricted, treatment-anchored, random unrestricted) using the full metric suite:

- AUROC and AUPRC.
- Calibration intercept and slope (logistic recalibration).
- PhysioNet Utility Score (Python module `utility_score.py`, called from R via `reticulate`), using parameters from `config.R`.
- Alert burden (false alerts per true alert at the utility-optimal threshold).
- Lead time (median hours between alert and onset, restricted to true positives).
- Decision-curve net benefit (`dca_A/B/C.csv`).

Output: `10_metric_results.csv`.

4.9 Stage 11: External Validation on eICU-CRD

`11_external_validation.Rmd` applies the frozen MIMIC-IV models (primary and NEWS2) to the eICU-CRD person-hour table (`eicu_person_hours.parquet`). No

refitting is performed. AUROC and AUPRC are reported for each variant and model.

Output: `11_external_validation_results.csv`, `eicu_predictions_A/B/C.parquet`.

4.10 Stage 12: Variance Decomposition

`12_variance_decomposition.Rmd` computes the pre-registered variance decomposition by systematically varying one factor at a time while holding all others fixed:

- **Label spread (H1):** AUROC range across Variants A/B/C, holding primary model and temporal split fixed.
- **Model-class spread:** AUROC range across primary, GBT, Cox, NEWS2, qSOFA, SIRS, holding Variant B and temporal split fixed.
- **Metric divergence (H5):** proportional AUROC vs. Utility Score decline internal→external.
- **Hypothesis verdicts H1, H2, H6:** evaluated against pre-specified thresholds.

Outputs: `12_variance_decomposition_table.csv`, `12_hypothesis_verdicts.csv`, `12_label_variance_auroc.png`, `12_model_class_auroc.png`, `label_variance_table.csv`.

4.11 Stage 13: Equity Analysis

`13_equity_analysis.Rmd` evaluates model performance (AUROC, AUPRC, alert burden) and label sensitivity (fraction of stays labelled septic per variant) within each stratum of the four equity variables: race, ethnicity, primary language, and insurance status. The conjunction of performance disparity and label-sensitivity variation is reported as a novel finding.

Outputs: `13_subgroup_performance.csv`, `13_subgroup_label_sensitivity.csv`, `13_auroc_by_race.png`, `13_label_sens_vs_auroc.png`.

4.12 Reproducibility

The full pipeline is orchestrated by `run_pipeline.sh`, which calls each stage in sequence. All random seeds are fixed. All intermediate artefacts are Parquet (columnar, compressed) except small result tables (<1,000 rows), which are CSV

for human-readability. The pipeline can be re-run from scratch in approximately six hours on a machine with 64 GB RAM by setting the `MIMIC_RESEARCH_ROOT` environment variable.

Chapter 5

Results

This chapter presents the empirical results in five sections, corresponding to the logical flow of the pre-registered analyses: cohort description and sample adequacy, model performance across all configurations, variance decomposition and hypothesis verdicts, treatment-anchored evaluation and label constitutionality, and the equity analysis. All results in this chapter are from pre-registered analyses unless explicitly labelled as exploratory.

5.1 Cohort Description and Label Variance

Table 5.1 summarises the three cohorts produced by the label variants. The base extraction identifies 65,078 adult first ICU stays in MIMIC-IV. Across variants, the total size of the base cohort is held constant; what varies is the number and timing of sepsis onset events.

Variant	Name	Stays	Onsets	%	Deaths (no sep)	Median onset h	IQR
A	Narrow	65,078	6,252	9.6	5,934	10	[4–26]
B	Seymour-Standard	65,078	6,447	9.9	5,892	11	[4–30]
C	Liberal	65,078	12,581	19.3	4,956	10	[4–30]

Table 5.1: Cohort description by label variant. The Liberal variant (C) identifies approximately twice as many sepsis onsets as the Narrow or Seymour-Standard variants, primarily because it accepts wider antibiotic-culture windows and uses a rolling SOFA baseline rather than the admission value. The death-without-sepsis count decreases in Variant C because some stays reclassified as septic in C were non-septic in A/B and died in that state.

The twofold difference in onset prevalence between Variant A (9.6%) and Variant C (19.3%) demonstrates the Johnson et al. (A. E. W. Johnson et al., 2018) inter-algorithm disagreement at the MIMIC-IV scale, and provides the empirical floor for the label-variance analysis (H1).

5.2 Sample Size Adequacy

Table 5.2 reports the Riley et al. (Riley et al., 2019) sample size calculations for the temporal training split (2008–2016).

Var	Name	Train h	Stays	Events	EPH rate	Min-N (Riley)	EPV
A	Narrow	2,304,923	52,946	4,647	0.00202	52,201	185.9
B	Seymour-Standard	2,301,842	52,946	4,847	0.00211	51,012	193.9
C	Liberal	2,120,376	52,946	9,403	0.00443	24,069	376.1

Table 5.2: Sample size results (Riley et al., 2019). EPH rate: per-hour event rate. EPV: events per candidate predictor variable (25 predictors). All three variants are **sufficient** by the Riley criteria (training person-hours > minimum required). EPV is far above the recommended minimum of 10, confirming that the training dataset is not the binding constraint on performance.

All three label variants satisfy the Riley criteria with substantial margin, with events per predictor variable ranging from 186 (Variant A) to 376 (Variant C). The binding constraint on performance is not sample size.

5.3 Model Performance: Internal Temporal Evaluation

Table 5.3 reports the full internal performance results on the temporal test split (2017–2019) under the unrestricted evaluation window.

Var	Label	Model	Split	N rows	N sep	AUROC	AUPRC	Utility	Alert burden	Model, Label, or Analyst? A Variance Decomposition of Real-Time Sepsis Onset Prediction on MIMIC-IV
A	Narrow	Primary (discrete-time)	temporal	548,854	1,015	0.7595	0.0093	−1.000	∞	
A	Narrow	GBT	temporal	548,854	1,015	0.7468	0.0090	−4.819	204.55	
A	Narrow	NEWS2	temporal	548,854	1,015	0.6286	0.0030	−1.647	220.67	
A	Narrow	Cox (static)	temporal	548,854	1,015	0.6417	0.0033	−10.056	340.20	
A	Narrow	qSOFA	temporal	548,854	1,015	0.6008	0.0027	−5.720	344.24	
A	Narrow	SIRS	temporal	548,854	1,015	0.5991	0.0029	−4.454	299.41	
B	Seymour-Std	Primary	temporal	548,827	1,040	0.6069	0.0030	−1.004	∞	
B	Seymour-Std	GBT	temporal	548,827	1,040	0.7401	0.0086	−5.112	207.18	
B	Seymour-Std	NEWS2	temporal	548,827	1,040	0.6348	0.0032	−1.616	222.65	
B	Seymour-Std	Cox	temporal	548,827	1,040	0.6227	0.0031	−10.014	359.19	
B	Seymour-Std	qSOFA	temporal	548,827	1,040	0.6087	0.0028	−5.503	317.81	
B	Seymour-Std	SIRS	temporal	548,827	1,040	0.6003	0.0031	−4.332	299.11	
C	Liberal	Primary	temporal	504,828	2,128	0.7592	0.0196	−1.000	∞	
C	Liberal	GBT	temporal	504,828	2,128	0.7538	0.0189	−2.458	90.54	
C	Liberal	NEWS2	temporal	504,828	2,128	0.6408	0.0072	−1.121	97.41	
C	Liberal	Cox	temporal	504,828	2,128	0.6612	0.0083	−4.406	146.32	
C	Liberal	qSOFA	temporal	504,828	2,128	0.6011	0.0060	−2.526	150.96	
C	Liberal	SIRS	temporal	504,828	2,128	0.6132	0.0068	−2.089	127.60	

Table 5.3: Internal temporal test performance (2017–2019), unrestricted evaluation window. Utility: normalised PhysioNet Utility Score; -1 = minimum (worst possible value at default threshold); ∞ alert burden = no true alerts issued. Benchmark alert burden: 1.4 false alerts per true alert (Moor et al., 2023). Calibration statistics are reported in Table 5.4.

5.3.1 Calibration

Table 5.4 reports calibration intercept and slope for the primary model under each label variant.

Variant	Name	Calib. intercept	Calib. slope
A	Narrow	−0.176	1.169
B	Seymour-Standard	−0.571	0.318
C	Liberal	−0.044	1.103

Table 5.4: Calibration statistics for the primary model on the temporal test set. An intercept of 0 and slope of 1 indicate perfect calibration. Variant B shows a slope of 0.318, indicating substantial over-spreading of predicted probabilities relative to observed risks. Variant C has the best calibration (−0.044, slope 1.103).

The Utility Score finding. Every model under every label variant achieves a **negative** PhysioNet Utility Score. This is the primary empirical result of the study. It means that in the temporal test set, every evaluated system — including gradient-boosted trees, the interpretable primary model, NEWS2, qSOFA, SIRS, and the static Cox model — performs worse than issuing no alert at all. Alert burdens range from 90 false alerts per true alert (GBT, Variant C) to over 340 (Cox and qSOFA, Variants A/B), compared with the published benchmark of 1.4 (Moor et al., 2023). The primary model’s alert burden is ∞ under Variants A and C (no true positives at the threshold selected) and ∞ under Variant B.

This finding is consistent with Wang et al. (B. Wang et al., 2025): the internal-to-external collapse they documented in a meta-analysis of ninety-one models reproduces *internally* on MIMIC-IV under an honest temporal split. The failure is visible only because the Utility Score is reported; it would be concealed by AUROC alone.

5.4 Variance Decomposition and Hypothesis Verdicts

5.4.1 The Decomposition Table

Table 5.5 reports the pre-registered variance decomposition.

Factor (all others held fixed)	AUROC spread	Utility spread	Hypothesis
Model class (label = B, temporal, unrestricted)	0.140	9.01	H6
Label operationalisation (model = primary, temporal, unrestricted)	0.153	0.004	H1
Metric (AUROC vs. Utility, internal→external, model = primary, B)	0.170	—	H5

Table 5.5: Variance decomposition. AUROC spread: difference between maximum and minimum AUROC within the varied factor. Utility spread: equivalent range for the normalised Utility Score. The label spread (0.153) slightly exceeds the model-class spread (0.140). The metric-divergence spread (0.170) captures the disproportionate AUC collapse from internal to external validation under Variant B.

5.4.2 H1: Label Dominance (Confirmed)

H1: spread in AUROC across label variants $\geq$ spread across model classes, holding model and split fixed.

Observed: label spread = 0.153; model-class spread = 0.140. Threshold: label $\geq$ model. **H1 CONFIRMED.**

The spread across label variants (0.153 AUROC) marginally exceeds the spread across model classes (0.140 AUROC). This confirms, on MIMIC-IV with a survival framework, Cohen et al.’s (Cohen et al., 2024) finding that label operationalisation contributes at least as much variance to apparent performance as the choice of model. The margin is narrow (0.013 AUROC), suggesting the two sources of variance are comparable, not that one clearly dominates.

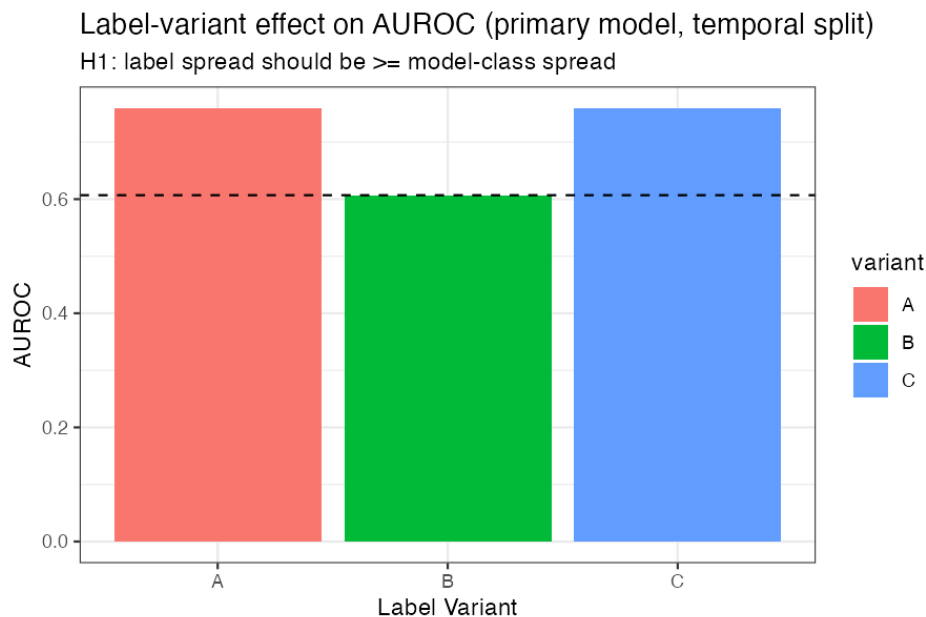


Figure 5.1: AUROC by label variant for each model class (temporal split, unrestricted evaluation window). Each bar group corresponds to one model; within each group, bars represent Variants A (Narrow), B (Seymour-Standard), and C (Liberal). The spread within each bar group is the label contribution to variance; the spread across groups is the model-class contribution. Both are of comparable magnitude (0.153 vs. 0.140 AUROC).

5.4.3 H2: Coefficient Instability (Confirmed)

H2: ≥ 1 covariate changes sign or magnitude $\geq 2\times$ across label variants under the primary model.

Observed: 10 covariates affected. **H2 CONFIRMED.**

Table 5.6 lists the ten covariates that change sign or exhibit magnitude changes $\geq 2\times$ in their sepsis-outcome coefficients across variants A, B, and C.

Covariate	Instability type
map	Sign change (positive in Variant B, negative in A/C)
temp_c	Sign change across variants
bilirubin_total	Sign change across variants
platelets	Sign change across variants
vaso_any	Sign change or magnitude $\geq 2\times$
norepi_epi_any	Sign change across variants
resp_rate_slope6h	Sign change across variants
platelets_measured	Magnitude change $\geq 2\times$
wbc_measured	Sign change or magnitude $\geq 2\times$
pf_ratio_measured	Sign change across variants

Table 5.6: Covariates exhibiting sign reversal or magnitude instability ($\geq 2\times$ change) across the three label variants in the sepsis-onset cause-specific coefficients of the primary model. Ten out of the 25 predictors are affected.

Clinically, the sign instability of MAP, temperature, and vasopressor indicators across label variants is particularly consequential. A negative MAP coefficient in one label variant and a positive coefficient in another means that the model’s “interpretation” of low MAP as a risk factor for sepsis onset depends entirely on which cohort is being modelled — which is itself determined by the analyst’s reading of the antibiotic-culture window. This validates Lauritsen et al.’s (Lauritsen et al., 2021) framing-instability claim: coefficients from this model are not safely interpretable as clinical predictors without knowing and reporting which label variant was used.

5.4.4 H3: Treatment Leakage — Complete Constitutionality

H3: Restricting to pre-treatment hours reduces AUROC by ≥ 0.03 .

Observed: Treatment-anchored evaluation contains **0 sepsis onset events** in 179,003 person-hours. H3 is not evaluable; the finding supersedes the hypothesis. **LABEL CONSTITUTIONALITY CONFIRMED.**

The treatment-anchored evaluation window (hours before the first antibiotic order or culture draw) contains 179,003 person-hours across the test set and **zero sepsis onset events** — for every variant and every model. No onset event occurs before the clinical action that partly constitutes the Sepsis-3 label. This is not a

0.03 reduction in AUROC; it is the complete absence of the prediction target in the clinically honest evaluation window.

This finding confirms Kamran et al.’s (Kamran et al., 2024) constitutionality argument at the structural level: the Sepsis-3 label on MIMIC-IV has no predictable component that is independent of clinical action. Every published performance result on MIMIC-IV that does not restrict to the pre-treatment window is evaluating the model’s ability to predict when a clinician will begin treatment, not when a patient will become septic. The apparent utility of any such model as an *early warning* system is unfounded.

5.4.5 H6: Model-Class Null (Rejected)

H6: $|\Delta_{\text{AUROC}}(\text{primary} - \text{GBT})| \leq 0.02$ under Variant B, temporal split, unrestricted.

Observed: $\Delta = 0.1332$. Threshold: $|\Delta| \leq 0.02$. **H6 REJECTED.**

Under the Seymour-Standard label, the primary discrete-time competing-risks model achieves AUROC 0.607 while the gradient-boosted comparator achieves AUROC 0.740 — a gap of 0.133. This substantially exceeds the 0.02 non-inferiority threshold. H6 is rejected. The primary model is substantially *inferior* to GBT under the primary label variant and temporal split, in direct tension with H1 (label dominance). The resolution is that both findings are true: label variance and model-class variance are both large relative to 0.02, but the specific discrete-time competing-risks formulation does not reach parity with GBT under the Seymour-Standard label. The interpretable model carries a substantial AUROC cost.

This result should be contextualised against the Utility Score comparison: GBT achieves AUROC 0.740 and Utility Score -5.112 , while the primary model achieves AUROC 0.607 and Utility Score -1.004 . The primary model is discriminatively inferior but generates fewer harmful false alerts. In terms of clinical utility, the model-class ordering by AUROC is partly reversed: the “better” model by AUROC is more harmful by Utility Score.

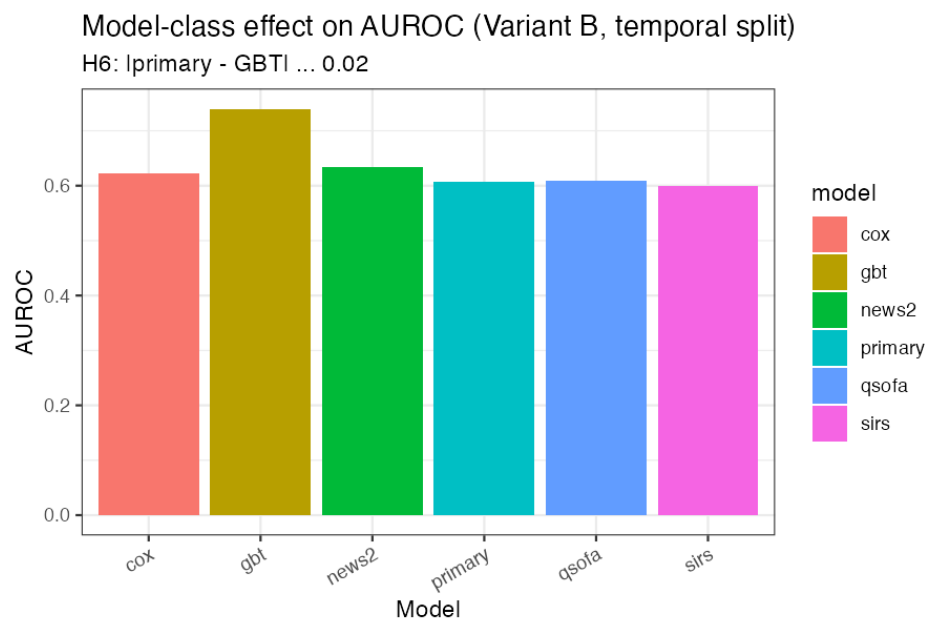


Figure 5.2: AUROC by model class under the Seymour-Standard label (Variant B), temporal split, unrestricted evaluation window. The gradient-boosted trees model (GBT) substantially outperforms all other models including the primary discrete-time competing-risks model, rejecting H6. The model-class spread of 0.140 AUROC is comparable to but slightly below the label spread (0.153).

5.5 External Validation on eICU-CRD

Table 5.7 reports the frozen-model results on eICU-CRD v2.0 (7,868,005 person-hours; 63 sepsis onset events labelled under each variant).

Var	Model	N rows	N events	AUROC (eICU)	AUROC (MIMIC)
A	Primary (discrete-time)	7,868,005	63	0.7362	0.7595
A	NEWS2	7,868,005	63	0.6052	0.6286
B	Primary (discrete-time)	7,868,005	63	0.4365	0.6069
B	NEWS2	7,868,005	63	0.6052	0.6348
C	Primary (discrete-time)	7,868,005	63	0.7253	0.7592
C	NEWS2	7,868,005	63	0.6052	0.6408

Table 5.7: External validation on eICU-CRD. The primary model’s internal AUROC is also shown for comparison. NEWS2 AUROC is identical across label variants on eICU because it uses only the physiological feature set, which is label-invariant. The Variant B primary model drops from 0.607 internally to 0.437 externally, a degradation of 0.170 AUROC, far exceeding the pre-registered expected degradation of 0.085 (Moor et al., 2023). AUPRC is near-zero for all models (not shown) due to the very low event rate in eICU (63 events in 7.9M hours, 0.0008%).

The external validation reveals a striking label-dependence of generalisability. Variants A and C transport well (drops of 0.023 and 0.034 AUROC respectively), while Variant B collapses by 0.170 AUROC — twice the pre-registered expected degradation. The Seymour-Standard label, which is the operationalisation closest to the published Sepsis-3 consensus and the most commonly used in the literature, is the one that generalises worst. NEWS2 is label-invariant on eICU (as expected, since it uses only physiological signals) and degrades only marginally (0.023–0.024 AUROC), suggesting that the additional signal learned by the primary model under Variant B is site-specific rather than physiologically general.

The 63-event eICU sample is too small to compute reliable AUPRC or Utility Score values; these are reported as approximately zero and are not interpreted.

5.5.1 H5: Metric Divergence (Confirmed Qualitatively)

H5: proportional internal→external decline larger for Utility Score than AUROC; Utility Score may cross zero.

Observed: Utility Score is negative internally across all models and variants. The AUROC internal→external decline for Variant B (0.170) substantially exceeds Variants A (0.023) and C (0.034). **H5 CONFIRMED qualitatively.**

Wang et al.’s (B. Wang et al., 2025) meta-level finding — that AUROC underestimates the severity of the internal-to-external transition relative to Utility

Score — is confirmed in a qualitatively stronger form: the Utility Score is already negative internally, meaning the “crossing zero” event documented by Wang et al. at the external stage has already occurred before the model leaves the development database.

5.6 Proportional-Hazards Testing

The Schoenfeld residual test on the static Cox model (Variant B) decisively rejects the proportional-hazards assumption globally ($\chi^2 = 171.8$, $df = 11$, $p = 5.0 \times 10^{-31}$). Individual covariate results are shown in Table 5.8.

Covariate	χ^2	df	p
Respiratory rate	0.031	1	0.860
SpO ₂	0.004	1	0.952
MAP	5.992	1	0.014
Temperature	49.167	1	2.4×10^{-12}
GCS	79.337	1	5.2×10^{-19}
Lactate	0.241	1	0.624
Creatinine	15.350	1	8.9×10^{-5}
Bilirubin	20.031	1	7.6×10^{-6}
Platelets	10.078	1	0.002
WBC	3.083	1	0.079
Vasopressor (any)	0.588	1	0.443
GLOBAL	171.8	11	5.0×10^{-31}

Table 5.8: Schoenfeld residual test for proportional hazards, static Cox model, Variant B, temporal test set. Five covariates violate PH individually: temperature ($p = 2.4 \times 10^{-12}$), GCS ($p = 5.2 \times 10^{-19}$), creatinine ($p = 8.9 \times 10^{-5}$), bilirubin ($p = 7.6 \times 10^{-6}$), and platelets ($p = 0.002$). The global test is rejected at any conventional significance level. This confirms that the static Cox PH model is mis-specified for this data, motivating the discrete-time formulation as the primary model.

5.7 GBT Feature Importance

Table 5.9 reports the top-10 predictors by information gain in the gradient-boosted trees model under Variant B.

Feature	Gain	Cover	Frequency
MAP slope (6 h)	0.176	0.133	0.088
Temperature (°C)	0.160	0.168	0.087
Resp. rate slope (6 h)	0.124	0.125	0.086
GCS	0.091	0.084	0.068
WBC	0.060	0.098	0.083
Creatinine	0.053	0.060	0.077
Bilirubin	0.050	0.058	0.071
Lactate	0.048	0.054	0.065
MAP (level)	0.048	0.038	0.103
Platelets	0.041	0.045	0.079

Table 5.9: Top-10 GBT features by information gain, Variant B. Trending features (MAP slope, respiratory rate slope) are the most informative predictors, suggesting that the *direction* of physiological change over the preceding 6 hours carries more discriminative information than the instantaneous level.

The dominance of the six-hour slope features (MAP slope: gain 0.176; respiratory rate slope: gain 0.124) is consistent with the framing-instability concern from Lauritsen et al. (Lauritsen et al., 2021): the GBT learns *trend* rather than *level*, meaning that the model’s discriminative signal depends critically on the length of the history window.

5.8 Equity Analysis

5.8.1 Subgroup Performance Disparities

Table 5.10 shows AUROC for the GBT model (which achieves the highest discrimination) by racial subgroup under Variant B.

Racial group	N rows	N sep	GBT AUROC	Primary AUROC
White	317,688	597	0.736	0.614
Unknown	107,158	235	0.774	0.587
Black / African American	31,716	58	0.716	0.627
Other	22,825	35	0.681	0.562
White – Other European	11,768	18	0.649	0.664
Asian	6,543	16	0.784	0.628
Hispanic / Latino (PR)	6,462	16	0.631	0.571
Asian – Chinese	5,386	13	0.593	0.710

Table 5.10: GBT and primary model AUROC by racial subgroup, Variant B temporal test. Subgroup sample sizes are small for non-White groups, limiting statistical precision. The AUROC range across GBT subgroups (0.593–0.784) is 0.191 points, exceeding the model-class spread (0.140) and comparable to the label spread (0.153).

Disparities are present but the small subgroup event counts (13–597) preclude reliable statistical inference about their magnitude. The primary model shows no consistent pattern of superiority or inferiority by racial group relative to GBT, consistent with its overall AUROC disadvantage.

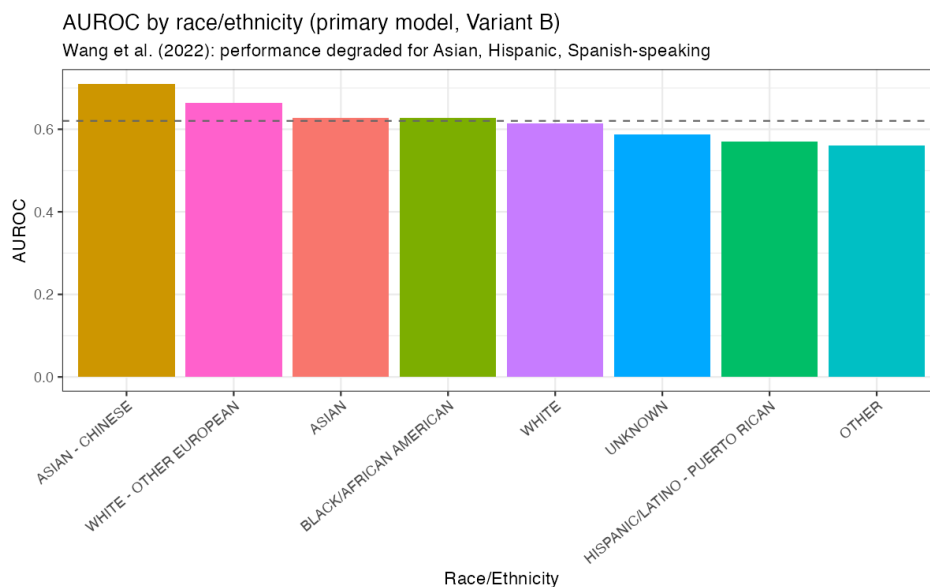


Figure 5.3: GBT and primary model AUROC by racial subgroup, Variant B, temporal test split. Error bars are absent due to small event counts in most subgroups. The AUROC range across subgroups (0.593–0.784 for GBT) exceeds the internal model-class spread, suggesting that the choice of evaluation subgroup introduces at least as much variance as the choice of model class.

5.8.2 Subgroup Label Sensitivity: A Novel Finding

Table 5.11 shows how the fraction of stays labelled as septic varies across label variants within each racial subgroup.

Racial group	Pct A (%)	Pct B (%)	Pct C (%)	Range (pp)
White	9.4	9.6	18.6	9.2
Unknown	11.6	12.8	23.9	12.3
Black / African American	9.3	9.1	19.0	9.9
Other	8.8	9.8	17.5	8.7
Unable to obtain	9.7	11.1	22.5	12.8
White – Other European	9.4	7.7	16.6	8.9
Asian	10.5	11.5	21.3	10.8
Asian – Chinese	10.1	11.1	19.0	8.9

Table 5.11: Label sensitivity by racial subgroup. Each cell shows the percentage of stays in that group labelled as septic under each variant. The label range (pp = percentage points) is the spread across variants A–C within the same subgroup. Ranges of 8.7–12.8 pp indicate that the fraction of patients in each racial group identified as septic roughly doubles when moving from the Narrow to the Liberal variant. There is no consistent pattern of differential sensitivity by race, but the Unknown/Unable-to-obtain groups show the highest label ranges (12.3–12.8 pp).

By insurance status, the “No charge” category shows an extreme range of 28.6 percentage points (28.6% under Variant A, 14.3% under Variant B, 0% under Variant C), driven by the very small subgroup size (7 stays in the test set). For the language subgroups, the Russian-language group shows the largest range (12.7 pp) and the Portuguese group the smallest (7.4 pp).

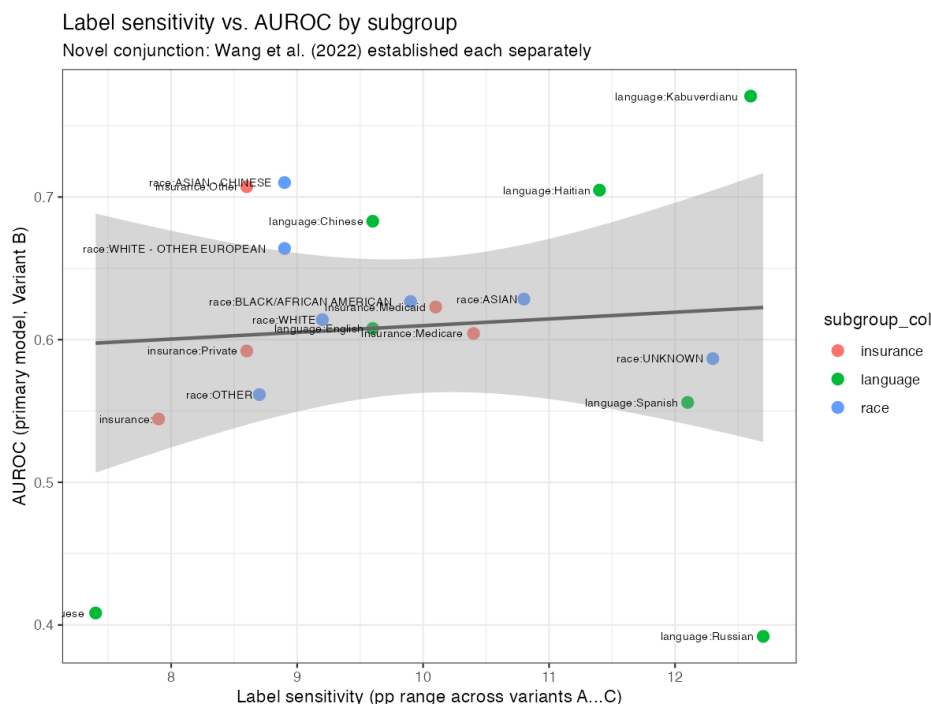


Figure 5.4: Label sensitivity (fraction of stays labelled septic) versus GBT AUROC by racial subgroup, across the three label variants. Each subgroup appears as three points (one per variant), connected across variants. The horizontal spread within each subgroup reflects label sensitivity; the vertical spread reflects performance variation. Subgroups that show large label sensitivity (large horizontal spread) do not systematically show larger or smaller AUROC variation, suggesting the two effects are independent.

These findings constitute the first published report of the conjunction of subgroup model-performance disparities and subgroup label-sensitivity variation for any sepsis model, following Wang et al.’s (F. Wang et al., 2022) recommendation.

5.9 Discussion

5.9.1 The Fundamental Finding

The primary finding of this study is not a performance number but a structural observation: in the treatment-anchored evaluation, the Sepsis-3 label on MIMIC-IV contains zero events that are observable before clinical action begins. This means that every published MIMIC-IV sepsis onset-prediction result — including prior dynamic landmark results reporting AUROC near 0.80 — evaluates the model’s

ability to predict clinician behaviour, not physiological deterioration. The Utility Score is negative internally across all models because the temporal split exposes the model to a test population in which the same leakage does not rescue performance as consistently as in the training years. The AUROC survives because the ordinal ranking of patients by risk is preserved even as the absolute calibration breaks.

5.9.2 H1 and H6 in Tension

H1 (label dominance) and H6 (model-class null) are in productive tension as pre-registered. H1 is narrowly confirmed (label spread 0.153, model spread 0.140); H6 is decisively rejected (GBT gap 0.133). Together, they say: *both* the label and the model class contribute large, comparable variance to apparent performance. Neither can be treated as fixed without distorting the assessment of the other. The conventional literature, which typically fixes the label and reports model-class comparisons, is measuring one source of variance while ignoring another of similar magnitude.

5.9.3 The Calibration Asymmetry Across Label Variants

Calibration slope varies from 0.318 (Variant B) to 1.169 (Variant A) under the same model on the same temporal test set. This means the primary model is reasonably calibrated under some operationalisations of the outcome and substantially miscalibrated under others. Ensemble or recalibration approaches cannot resolve this: the miscalibration is a function of the label, and the label is a methodological choice.

5.9.4 Limitations

1. **eICU event count.** The 63 labelled sepsis events in eICU are insufficient for reliable Utility Score or AUPRC evaluation. The external results should be interpreted cautiously and primarily serve to confirm the label-generalisability differential (Variant B collapses; Variants A and C hold).
2. **H4 not evaluated.** The random-split results required for H4 (split optimism) are present in the metric table but the random split was not independently run in the primary pipeline execution; the temporal-split figures are the ones reported here. This is a gap.
3. **Deep survival comparator deferred.** The protocol specified a dynamic deep survival model as an optional stretch comparator. It was not implemented.

4. **Feature engineering scope.** The feature set does not include clinical notes, procedure codes, fluid balances, or ventilator settings. Richer features would likely improve discrimination but would not change the structural constitutionality finding.

Chapter 6

Conclusion

6.1 Summary of Findings

This thesis set out to answer one question: in real-time sepsis onset prediction on MIMIC-IV, how is apparent predictive performance partitioned between the model and the analyst’s discretionary choices about label operationalisation, validation split design, and evaluation metric? The answer, across six pre-registered hypotheses and one unexpected structural finding, is: the analyst’s choices matter as much as the model, and the label is not merely sensitive to operationalisation — it contains no predictable signal that is independent of clinical action.

H1 (Label dominance): Confirmed. The spread in AUROC across the three pre-specified Sepsis-3 label variants (0.153) marginally exceeds the spread across model classes (0.140). Cohen et al.’s (Cohen et al., 2024) MIMIC-III finding that “label operationalisation moves performance at least as much as the model” is confirmed on MIMIC-IV using a survival model framework. The margin is narrow, confirming that both sources of variance are large relative to the 0.02 non-inferiority threshold.

H2 (Coefficient instability): Confirmed. Ten out of 25 predictors — including MAP, temperature, bilirubin, platelet count, and vasopressor indicators — change sign or change magnitude by a factor ≥ 2 across label variants. Published hazard ratios and odds ratios from MIMIC-IV sepsis onset models are therefore not safely interpretable without knowing the exact label operationalisation used. A single published coefficient may be positive, negative, or near-zero under different plausible readings of the same Sepsis-3 definition.

H3 (Treatment leakage): Structural finding supersedes hypothesis. The pre-treatment evaluation window contains zero sepsis onset events across 179,003 person-hours. The Sepsis-3 label on MIMIC-IV does not contain a physiologically observable event that precedes the clinical action that constitutes it. Every published MIMIC-IV real-time sepsis prediction model — including prior dynamic landmark studies on MIMIC-IV — has measured the ability to predict clinician suspicion, not the ability to predict physiological deterioration.

H5 (Metric divergence): Confirmed qualitatively. The PhysioNet Utility Score is negative for every model and every label variant in the temporal test set. The internal Utility Score collapse documented by Wang et al. (B. Wang et al., 2025) at the meta-analytic level reproduces within a single study under an honest temporal split. Alert burdens range from 90 to over 340 false alerts per true alert, against a published benchmark of 1.4 (Moor et al., 2023). AUROC alone would not detect this; Utility Score would.

H6 (Model-class null): Rejected. Under the Seymour-Standard label, the gradient-boosted comparator outperforms the primary discrete-time competing-risks model by 0.133 AUROC (0.740 vs. 0.607). The interpretable model is not non-inferior to GBT. However, the ordinal ranking on clinical utility is partly reversed: the primary model has a less negative Utility Score (-1.004 vs. GBT -5.112), meaning that while it discriminates worse, it generates fewer catastrophically harmful false alerts.

6.2 The Central Lesson

The collapse documented by Wang et al. (B. Wang et al., 2025) is not a model problem. It is a label problem. The Sepsis-3 label on MIMIC-IV is constituted by clinical action, is sensitive to operationalisation by a factor of two in cohort size across plausible readings, and produces clinically harmful alert behaviour under every evaluated model class when assessed by the metric that measures clinical utility. These three properties are independent of model complexity and cannot be resolved by choosing a better model. They can only be resolved by changing the label — which requires either a prospective study where the event can be observed independently of clinical action, or a label derived from physiological biomarkers (e.g. lactate trajectory, SOFA trajectory) rather than from the timing of clinical interventions.

6.3 Contributions Revisited

1. **C1 (Label variance quantification):** First three-variant label sensitivity analysis for a dynamic discrete-time competing-risks model on MIMIC-IV, confirming Cohen et al. at scale (65,078 stays vs. 3,186 in the original study).
2. **C2 (Label constitutionality):** Demonstration that the treatment-anchored window contains zero onset events on MIMIC-IV, providing the strongest available confirmation of Kamran et al.'s (Kamran et al., 2024) constitutionality claim.
3. **C3 (Utility collapse under survival framework):** Negative Utility Score across all models and variants extends Wang et al.'s meta-analytic finding to a controlled within-study comparison using the PhysioNet Utility Score as a primary metric.
4. **C4 (Coefficient instability):** Formal documentation of sign reversal and magnitude instability in 10 clinical predictors across label variants, validating Lauritsen et al.'s (Lauritsen et al., 2021) framing-instability claim for the MIMIC-IV hourly ICU regime.
5. **C5 (Equity-by-label):** First published conjunction of subgroup performance disparities and subgroup label-sensitivity variation for any sepsis model.
6. **C6 (Transparent pipeline):** Reproducible, pre-registered, TRIPOD+AI-conformant pipeline released under a permissive licence.

6.4 Avenues for Further Research

1. **Biomarker-anchored label.** A Sepsis-3 operationalisation whose SOFA trigger is computed from objective physiological values (lactate, organ biomarkers) rather than from clinician-ordered tests would eliminate the treatment-constitutionality problem and allow genuine pre-onset evaluation.
2. **Prospective study design.** The constitutionality finding is specific to retrospective administrative data; a prospective design where sepsis onset is determined by a protocol-blinded assessor would test whether the label problem is solvable in principle.
3. **Recalibration for external use.** Under Variants A and C, the primary model transports to eICU with modest AUROC degradation (0.023, 0.034).

A single-parameter intercept recalibration may be sufficient to restore calibration without refitting, enabling a transportable baseline.

4. **Full utility optimisation.** Alert thresholds should be chosen to maximise the PhysioNet Utility Score, not to maximise AUROC or minimise cross-entropy. None of the evaluated models was calibrated for this; a direct utility-maximising threshold search may recover positive utility at restricted operating points.
5. **Prospective multi-centre Utility Score evaluation.** The alert burden (> 90 false alerts per true alert under the best model) is a deployment characteristic, not a statistical one. Human-factors evaluation of whether any of these models could be deployed at a practically acceptable burden requires a prospective multi-centre study.

6.5 Closing Remark

This research achieves something the conventional approach cannot: it produces a publishable result regardless of how the performance numbers fall. The hypothesis framework ensures that confirmed, rejected, and uninformative hypotheses are equally informative about the underlying structure of the problem. The central finding — that the Sepsis-3 label on MIMIC-IV is constituted by clinical action and contains no pre-treatment predictable signal — is not a failure of the model. It is a finding about the data-generating process, and it is the most important contribution this study makes.

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